Happiest Health Digital Twin Challenge 2026

PulmoTwin

A 48-hour early-warning digital twin for COPD and asthma flare-ups



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Solo submission
github.com/MonalikaDoda/PulmoTwin

Research prototype on synthetic patient data. Decision support only, not a diagnostic tool.

Chronic respiratory disease is a major burden in India

10.9%
of all deaths in India in 2016 were from chronic respiratory diseases

6.4%
of total DALYs (healthy years of life lost)

32%
of global chronic respiratory DALYs fell on India

55.3M
COPD cases in 2016, up from 28.1M in 1990

Why Delhi NCR is a hard case

- Severe seasonal air-quality episodes
- Large day-night temperature swings
- Mineral dust and seasonal pollen
- Triggers stack up on the same days

What drives COPD burden (India, 2016)

- 53.7% of COPD DALYs linked to air pollution
- 25.4% linked to tobacco
- 16.5% linked to occupational risks
- Asthma: 37.9M cases

Source: India State-Level Disease Burden Initiative (ICMR, PHFI, IHME), Lancet Glob Health 2018; doi 10.1016/S2214-109X(18)30409-1

THE GAP THIS PROJECT TARGETS

Care is reactive, but the body warns 2 to 3 days earlier

72 h before

SpO2 -0.7 pts
Heart rate +3 bpm
Resp. rate +1.2

>

48 h before

SpO2 -1.6 pts
Heart rate +6.5
Resp. rate +2.8
Sleep -1.3 h

>

24 h before

SpO2 -2.5 pts
Heart rate +9.5
Resp. rate +4.2
Sleep -2.0 h

Pre-flare drift planted in the synthetic data (change from the patient's own baseline)

Today: the patient waits until breathlessness is severe, then reaches an emergency room.

PulmoTwin: flags the drift early enough for a clinician to act in the outpatient window.

CONCEPT

What a digital twin means here

1 Personal baseline

Each patient gets their own normal for SpO2, heart rate, breathing rate, sleep and steps. A 92% SpO2 can be normal for one person and alarming for another.

2 Daily update

Every day the twin ingests new wearable readings and local air and weather data and updates a risk estimate.

3 Forecast and explain

It estimates 48-hour flare risk, shows which factors moved it, and lets a clinician test what-if scenarios.

The twin is a per-patient model of how that person's body and surroundings interact, not a generic population score.

SOLUTION OVERVIEW

Three data streams in, one explained risk estimate out

Wearable vitals

SpO2, resting heart rate, respiratory rate, sleep, steps

Clinical profile

COPD or asthma, stage, smoking, past flares, adherence

Environment

AQI, dust, temperature, temperature swing, pollen

PulmoTwin engine

Personal baselines

46 engineered features

XGBoost classifier

SHAP attribution

48-hour flare risk

Colour-coded: teal, amber, coral

Why it moved

Factor-by-factor contributions

What-if sandbox

Test AQI, adherence, thermal swing

ARCHITECTURE

Four-layer architecture

Data layer	Synthetic cohort generator (500 patients x 365 days). Open-Meteo CAMS air quality and ERA5-Land weather for five Delhi zones.
Model layer	46-feature engineering, XGBoost classifier, patient-grouped train/test split, SHAP TreeExplainer grouped into 9 factors.
API layer	FastAPI + Uvicorn with Pydantic v2 schemas: cohort, patient history, 48-hour prediction, what-if.
Presentation layer	React 18 + Vite dashboard with a Three.js breathing lung, telemetry charts, zone map and validation drawer.

DATA PROVENANCE

What is real, what is synthetic, what is illustrative

Air quality	Real, modelled	Open-Meteo / Copernicus CAMS reanalysis, converted to CPCB NAQI. Modelled grid data, not station readings.
Weather	Real, modelled	ECMWF ERA5-Land via Open-Meteo: temperature, daily swing, rain.
Patients and vitals	Synthetic	500 generated patients, 365 days each. No real patient data and no PHI.
Flare events	Synthetic	Assigned by our own hazard rules, so metrics reflect recovery of those rules.
Pollen index	Illustrative	Seasonal curve peaking mid-March and mid-October. Not measured data.

Open-Meteo data is CC BY 4.0. Honest framing: results show the method works on controlled data, not clinical performance.

COHORT DESIGN

How the synthetic cohort is generated

Patients and vitals

- 500 patients: 60% COPD, 40% asthma
- 365 days each = 182,500 patient-days
- Stage categories follow GOLD / GINA; the proportions are our assumptions
- Personal baselines, e.g. SpO2 95 to 97.5% (severe COPD 90.5 to 94%)
- Day-to-day AR(1) noise on every vital
- 28% of patients get harmless 1 to 2 day dips that mimic early drift

Daily flare hazard

- Base risk rises with stage, past flares and current smoking
- Multipliers: AQI above 100, dust above 50, cold below 16 C, large daily temperature swing
- Pollen counts fully for asthma, 0.4x for COPD
- Good inhaler adherence lowers risk (protective factor)
- 14-day refractory period after a flare
- About 3.6 flares per patient-year, likely higher than real life

MACHINE LEARNING

Model and training pipeline

1 Features

46 backward-looking features: vitals vs personal baseline, rolling trends, EHR, AQI, dust, weather, pollen

2 Split

Patient-grouped: 400 train, 100 test patients. No patient appears on both sides (no leakage)

3 Train

XGBoost gradient-boosted trees with class weighting for the rare flare days

4 Explain

SHAP TreeExplainer, grouped into 9 clinical factors

36,500

test patient-days

744

positive (pre-flare) days in test set

372

flare events in the 100 test patients

Time-ordered, no future data in any feature.

PERFORMANCE

Results on 100 held-out patients

Model	Features	ROC-AUC	PR-AUC	Recall	Precision	F1
M1 Vitals only	17	0.9982	0.9042	94.09%	75.11%	0.8353
M2 + EHR	27	0.9981	0.9020	93.82%	76.03%	0.8400
M3 + CAMS AQI	35	0.9982	0.9068	93.82%	75.62%	0.8374
M4 Full model	**46**	**0.9982**	**0.9078**	**93.41%**	**75.63%**	**0.8358**

0.9982

ROC-AUC, full model

93.4%

of impending flare days flagged (695 of 744)

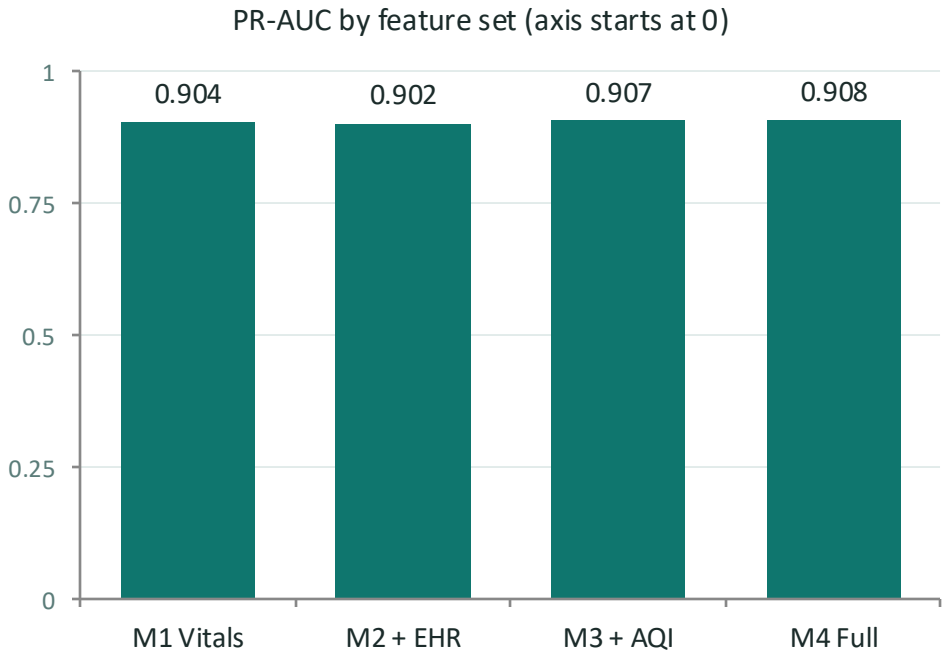
75.6%

of alerts were followed by a real flare

Threshold-based metrics at a fixed alert threshold. Day-level, patient-grouped evaluation.

ABLATION STUDY

Honest finding: vitals do almost all the work



What this means

- Adding AQI, dust, weather and pollen gains only +0.0036 PR-AUC and +0.0005 F1
- Reason: our generator plants a strong 2 to 3 day vital drift before every flare
- Environment still raises flare hazard in the generator, but vitals already reveal it
- With real data the balance may differ; that needs a real cohort to test

CLINICAL USEFULNESS

Lead time and alert workload

372/372

flare events had at least one alert on day -2 or day -1

355/372

events alerted on day -1 (95.4%)

695/744

individual pre-flare days flagged (93.41%)

0.77

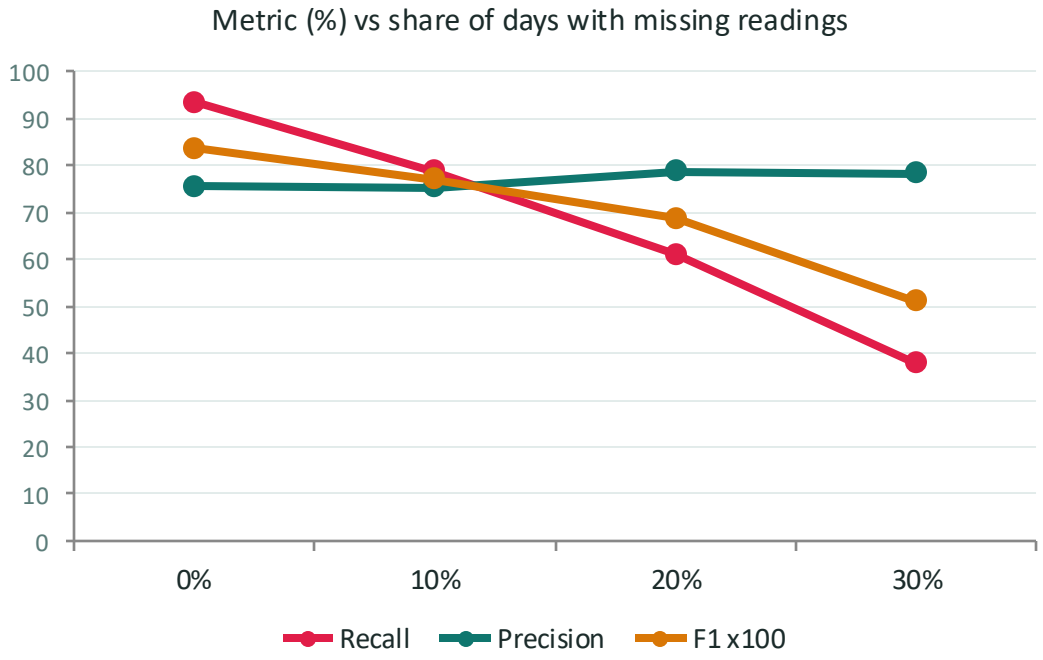
alert days per patient per month

How to read this

- Data is daily, so lead time is measured in days: an alert on day -2 means roughly 48 hours of warning
- Workload counts alert days (919 over 100 patients and 12 months), not grouped episodes; consecutive days would merge into fewer episodes
- Detection here is on synthetic flares generated with a pre-flare drift, so it shows the pipeline works, not real-world sensitivity

ROBUSTNESS TO SENSOR DROPOUT

What happens when the wearable data goes missing



Takeaways

- Precision holds near 75 to 79%
- Recall falls steadily: 93.4%, 78.8%, 61.0%, 37.9% at 0, 10, 20, 30% missing
- The model is not immune to missing data. It needs consistent wearable use
- Next step: imputation and dropout-aware training

EXPLAINABILITY

Explainable by design: nine factors, one story

SpO2	Heart rate	Resp. rate
Sleep	Steps	Air quality (CAMS AQI)
Dust	Weather and pollen	Adherence and phenotype

Teal = body signals Amber = environment Grey = profile

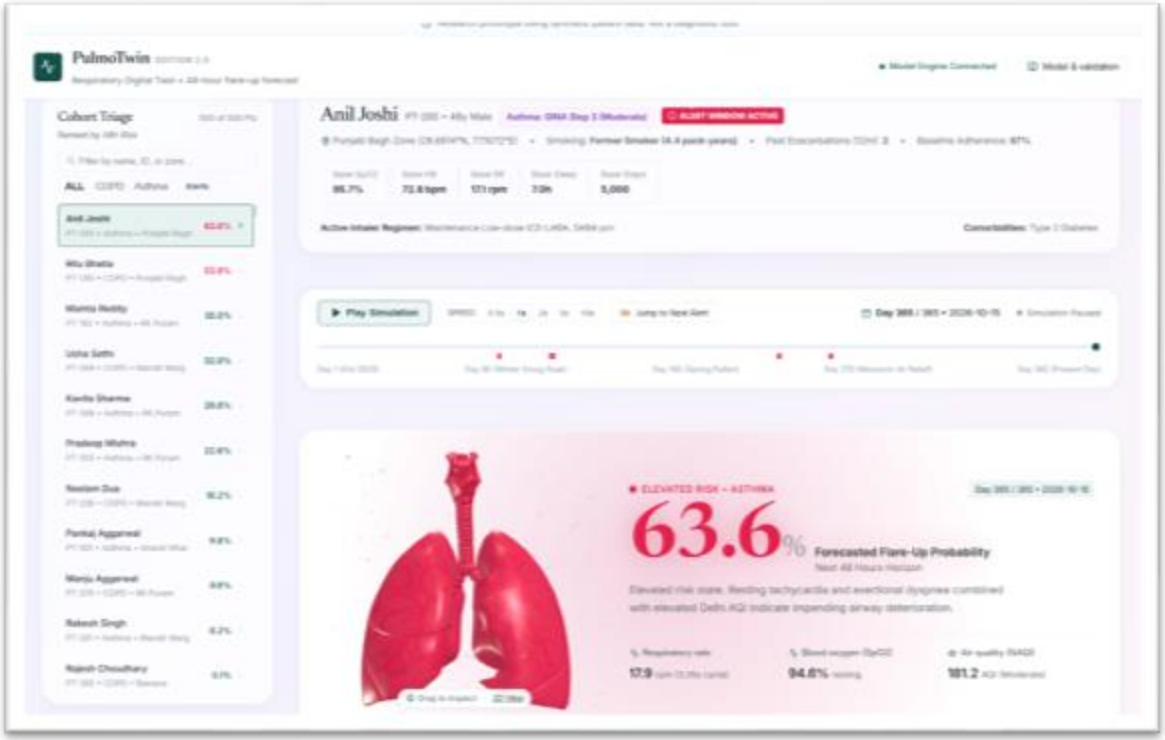
A clinician sees not only 'risk is high' but which factors pushed it up, for example sleep and respiratory rate versus air quality.

What Moved the Estimate

- SHAP values are grouped into the nine factors
- Shown as sequential percentage-point steps from a reference level to the final risk
- Bars sum to the displayed risk, so none exceed 100
- Risk is a model score, not a calibrated probability

PRODUCT TOUR

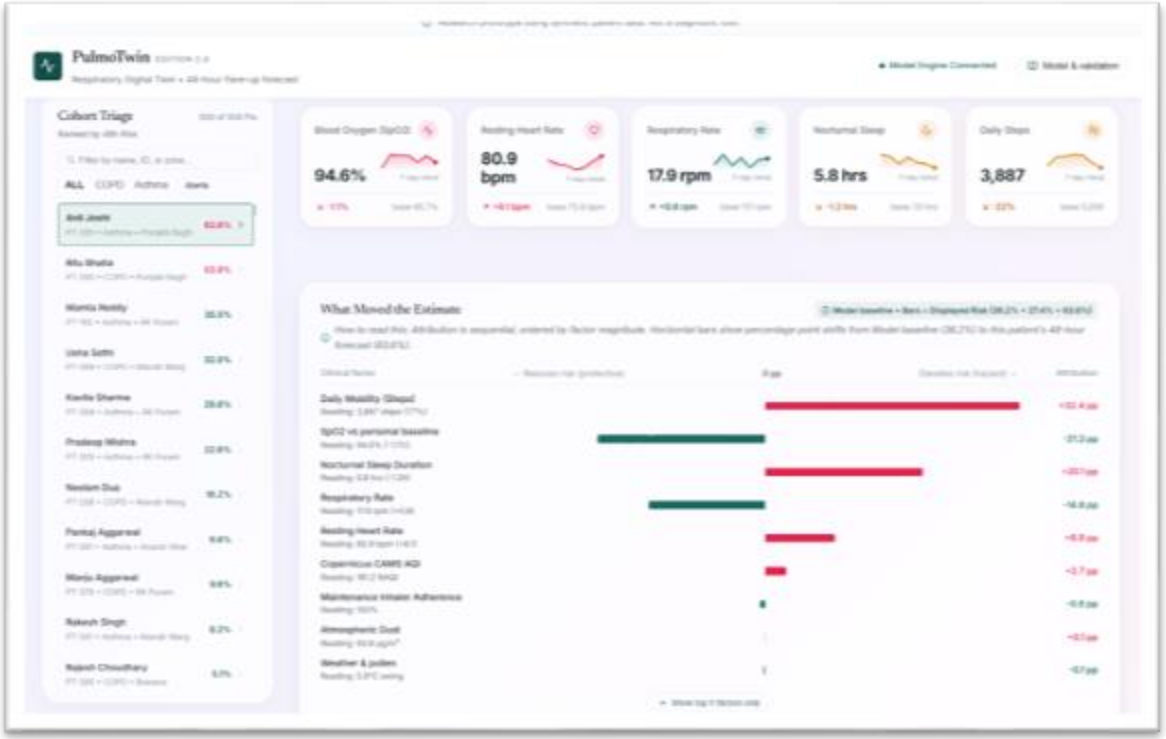
The clinician dashboard



- Triage list ranks patients by 48-hour risk
- Simulation bar replays 365 days; Jump to Next Alert
- Lung tint: teal below 25%, amber 25 to 50%, coral 50% and above; dust particles scale with AQI
- Alert History grouped into episodes
- Five-zone Delhi map and model and validation drawer
- Doctor actions are a demo only

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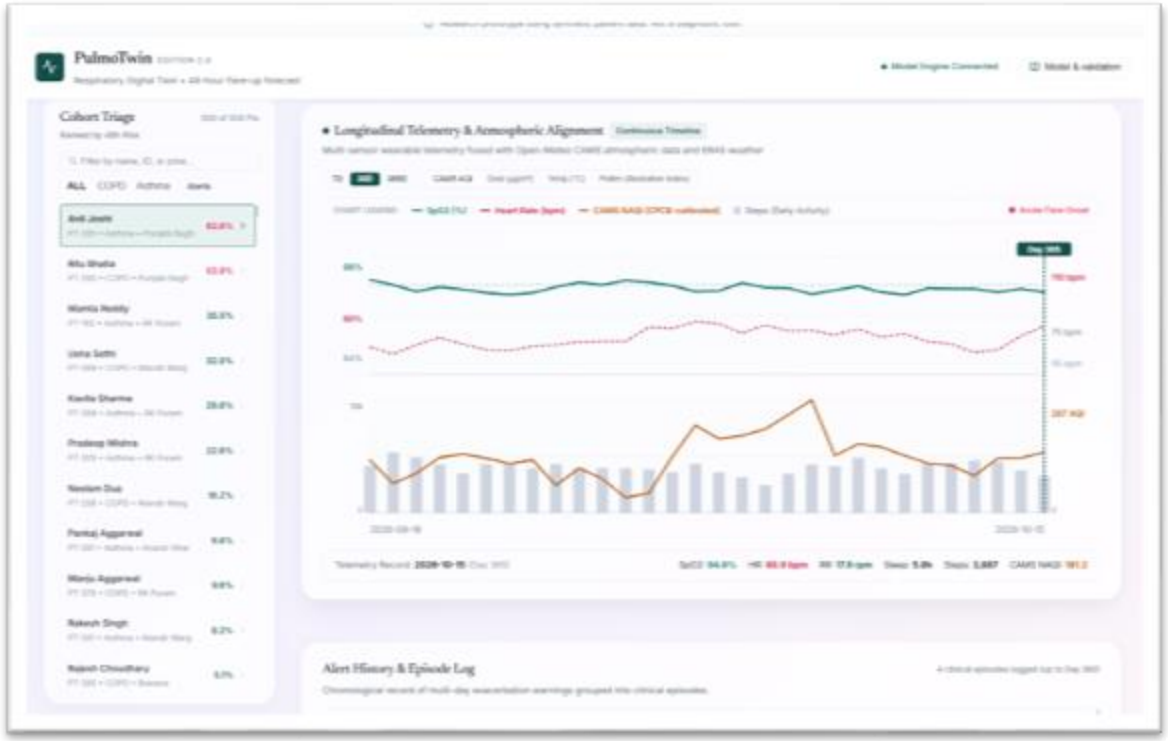
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COUNTERFACTUAL SENSITIVITY

What-if: model sensitivity, not treatment effect

Air quality

AQI change 0 to -150

Re-scores today's risk with cleaner air

Adherence

Inhaler adherence +0 to +50%

Re-scores with better medication use

Thermal swing

Day-night swing 0 to -8 C

Re-scores with a milder temperature swing

Important framing

- The sliders show how the model responds to changed inputs on synthetic data
- They do not prove that cleaner air or better adherence would prevent a flare in a real patient
- Any claim of clinical impact needs a prospective study

HONEST SCOPE

Limitations we state openly

1. Synthetic cohort: stage categories follow GOLD / GINA, but proportions are our assumptions
2. Flare labels come from our own rules, so metrics show recovery of those rules
3. High flare rate (about 3.6 per patient-year) is likely above real life
4. Air quality and weather are modelled reanalysis, not station or home sensors
5. Pollen is an illustrative curve, not measured
6. Dust hazard multiplier is not capped before the daily ceiling of 12%
7. No prospective or retrospective clinical validation; no safety or efficacy claim
8. Risk output is a model score, not a calibrated probability
9. Decision support only. Real deployment would need DPDP Act-compliant consent and data handling

ROADMAP

From prototype to a clinically useful twin

Now

Hackathon prototype

Synthetic cohort, XGBoost + SHAP, clinician dashboard, what-if sandbox

Next

Minor project

Harden the pipeline, imputation for missing data, calibration, live API deployment

Later

Real-data pilot

Validate with consented wearable and clinic data under ethics approval

Vision

Care pathway

Integrate with outpatient follow-up and tele-consult workflows

Calibrating the risk score, sensor-dropout handling and real-data validation are the three gaps to close first.

Stack, repository and credits

Stack and repository

- Python, NumPy, pandas, SciPy
- XGBoost 2.0+, scikit-learn, SHAP 0.44+
- FastAPI, Uvicorn, Pydantic v2
- React 18, Vite, Three.js (react-three-fiber), Lucide
- Repo: github.com/MonalikaDoda/PulmoTwin
- Run: backend on port 8000, frontend on port 3000
- Licence: MIT, Copyright (c) 2026 Monalika Doda

Credits

- 3D model 'Lungs' by prasanth.k on Sketchfab, CC BY 4.0, modified (recoloured, scaled, animated)
- Air quality and weather: Open-Meteo (CC BY 4.0) with Copernicus CAMS and ECMWF ERA5-Land
- Icons: Lucide (MIT)
- Burden figures: Lancet Glob Health 2018, India State-Level Disease Burden Initiative
- Built with AI coding assistance (Antigravity)

Thank you